HPA Project

House Price Prediction, Property Recommendation & Blockchain Agent

Demo overview — simple explanation for audience

What is this project?

This project combines real estate data, two machine learning models, blockchain operations, and an AI agent. You can talk to the agent in plain English — it figures out what you want and runs the right function (tool) for you.

Data we start with

- Raw property data from Gurgaon (10,000+ listings) with details like bedrooms, bathrooms, area, location, photos, price, society name, and more.
- We clean and prepare this data before training — for example: filter sale listings, convert property type to numbers, extract sector and GPS coordinates.

Model 1 — Price Prediction (XGBoost)

Goal: Given property features, predict an approximate price (MAX_PRICE).

14 input features (example):
Bedrooms, Bathrooms, Balconies, Facing, Age, Carpet sqft, Super built-up sqft, Property type (4 categories), Sector number, Latitude, Longitude

Target: MAX_PRICE
Model: XGBRegressor — saved as property_price_model.pkl
Accuracy: $R^2 \approx 0.91$ | Error (MAE) ≈ ± ■31.8 Lac

- The model learns which features matter most (e.g. area and location often weigh heavily).
- At runtime: pass 14 comma-separated numbers → get predicted price instantly.

Model 2 — Similar Property Recommendation (KNN)

Goal: Find properties in our database that are most similar to what the user wants.

- Uses K-Nearest Neighbors with Euclidean distance on the same 14 features.
- Location (latitude/longitude) is weighted more heavily so nearby areas rank higher.
- Returns the top 15 closest matches with a similarity score.
- Output: recommended_properties.csv + recommended_properties.pdf (one page per property).
- Interactive web viewer (viewer.py) to browse results with photos and details.

Blockchain — Read & Write (Web3)

Reusable functions connect to a smart contract on BSC testnet:

- balanceOf — check token balance for any wallet address (read).
- allowance — check how much a spender is allowed to use (read).
- transferFrom — move tokens from one address to another (write / transaction).

In a demo story: transfer can represent buying or paying for a property on-chain.

The AI Agent — how it ties everything together

An agent is a loop: LLM (Gemini) + tools (Python functions) + memory.

How the agent thinks (ReAct pattern):
1. Thought — understand what the user wants
2. Action — pick a tool (predict price, recommend, balance, transfer…)
3. Observation — read the tool result
4. Final Answer — reply in plain language

- Each tool has a description — the LLM reads it and picks the best match.
- You can ask in natural language: “predict price for ...”, “suggest similar properties ...”, “what is my

Suggested demo flow

- 1. Ask agent to predict price — paste 14 feature values.
- 2. Ask agent to recommend similar properties — same features → CSV + PDF.
- 3. Open viewer.py or share the PDF brochure with the audience.
- 4. Show blockchain read: balance / allowance for a wallet.
- 5. Show blockchain write: transfer tokens (property purchase story).

Scope for improvement

- Improve prediction accuracy with more data and tuning.
- Better agent memory for longer, smarter conversations.
- Add RAG — analyze locality scores, living quality, and facilities before final answer.
- Auto-invest agent — monitor news, invest in properties, flag scams.

Project folders (quick map)

- House_prediction_model/ — training notebook, predict.py, saved XGB model
- House_recommendation/ — KNN notebook, CSV/PDF output, viewer, recommendation.pkl
- ContractOperations/ — Web3 read/write (rw.py)
- AGENT/ — agent.py + tool.py (LLM + all tools)